Inventory Management System Project Report

Python, Java, and Data Analytics Implementation

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WEEK 5: June 1, 2026 – June 7, 2026

Project Overview

The Inventory Management System project focuses on designing and implementing an efficient software solution to manage stock levels, product tracking, forecasting, and reporting. The system integrates analytics tools to improve inventory decisions and operational efficiency.

Objectives

- Enhance the frontend dashboard with a professional and responsive user interface.
- Implement complete CRUD operations through frontend forms and backend APIs.
- Improve backend reliability with validation, error handling, and HTTP status codes.
- Integrate analytics and inventory statistics for better monitoring and decision-making.

Roles and Responsibilities

- Developed dynamic dashboard components to display products, suppliers, transactions, and low-stock alerts.
- Implemented frontend forms and connected them with backend CRUD APIs.
- Enhanced backend services with validation, filtering, search functionality, and improved SQLite integration.
- Performed API testing, frontend-backend integration testing, and updated project documentation.

Tools and Technologies Used

Python, Java, Spring Boot, SQLite, SQL, Pandas, Matplotlib, GitHub, VS Code, Postman

Key Outcomes

- Successfully integrated frontend and backend components into a functional inventory management application.
- Implemented analytics dashboards, CRUD forms, and inventory statistics features.
- Improved application usability, data validation, and overall system reliability.

Conclusion

Week 5 focused on transforming the project into a production-style inventory management system with complete frontend integration and enhanced backend functionality. The completed features improved usability, analytics, and overall system performance while preparing the application for future deployment and scalability.

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